
Large Language Models

Large language models

Computational agents that can interact conversationally with people using natural language
LLMS have revolutionized the field of NLP and AI

Fundamental intuition of large language models

Text contains enormous amounts of knowledge

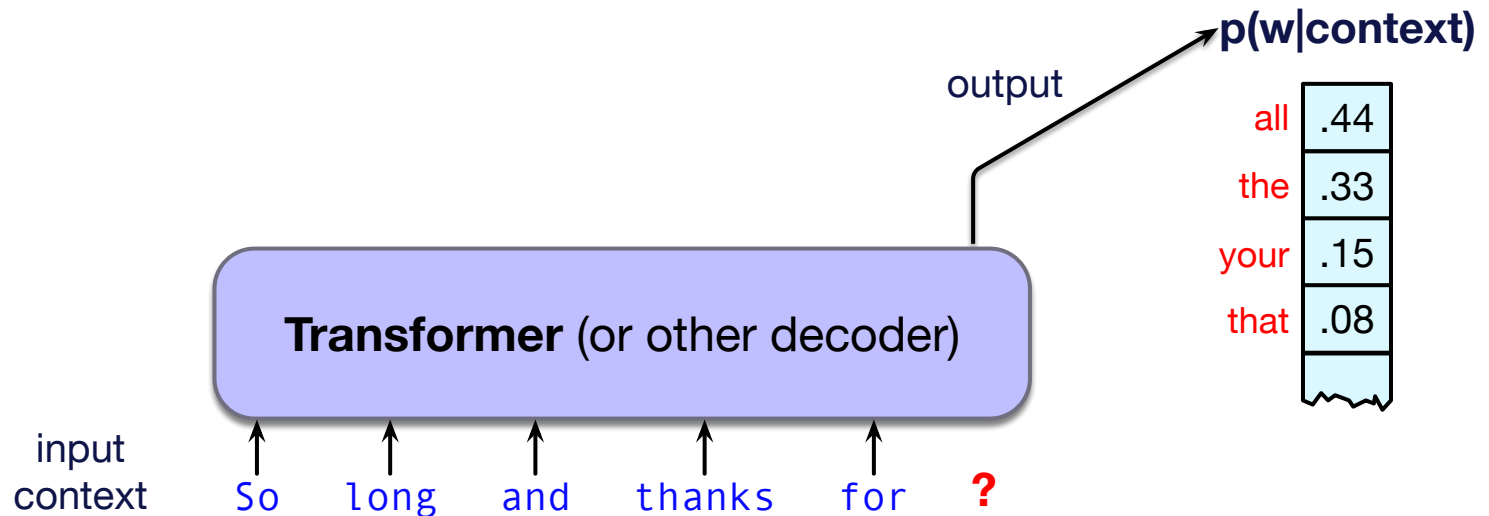
Pretraining on lots of text with all that knowledge is what gives language models their ability to do so much

What is a large language model?

A neural network with:

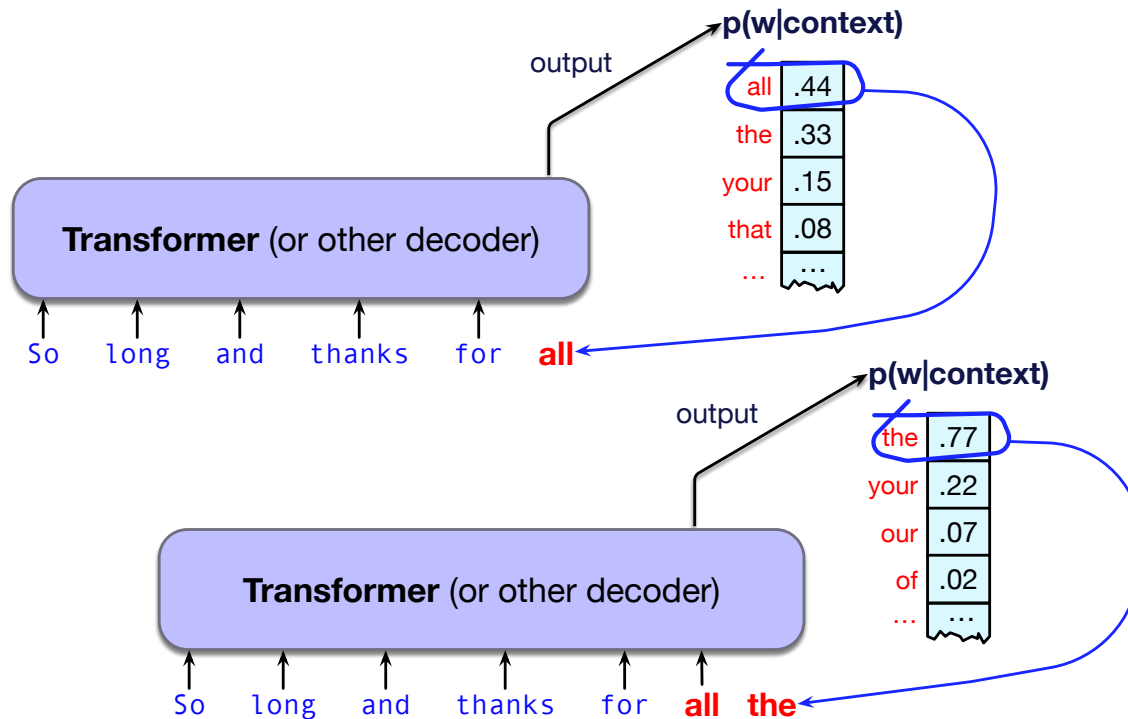
Input: a context or prefix,

Output: a distribution over possible next words



LLMs can generate!

A model that gives a probability distribution over next words can generate by repeatedly sampling from the distribution



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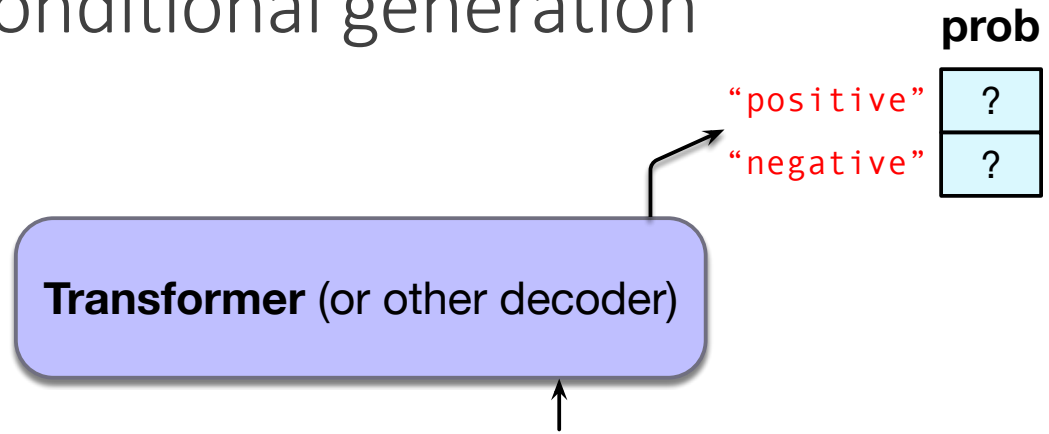
Conditional Generation of Text: The Intuition



Big idea

Many tasks can be turned into tasks of predicting words!

Sentiment via conditional generation



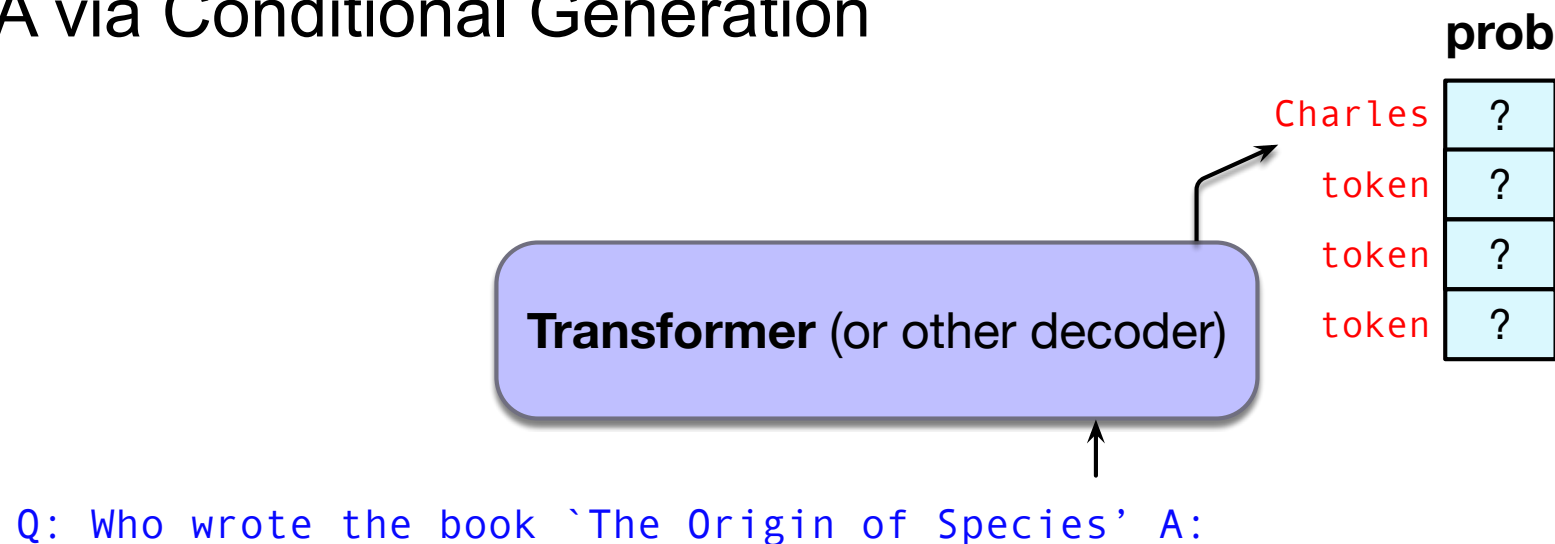
The sentiment of the sentence "I like Jackie Chan" is:

Which word has a higher probability?

$P(\text{positive} | \text{The sentiment of the sentence "I like Jackie Chan" is:})$

$P(\text{negative} | \text{The sentiment of the sentence "I like Jackie Chan" is:})$

Q/A via Conditional Generation



Now we iterate:

$P(w|Q: \text{Who wrote the book "The Origin of Species"}? \text{ A: Charles})$

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Prompting

Prompt

Prompt: a text string that a user issues to a language model to get the model to do something useful by conditional generation

Prompt engineering: the process of finding effective prompts for a task.

Prompts

A question:

What is a transformer network?

Perhaps structured:

Q: What is a transformer network? A:

Or an instruction:

Translate the following sentence into Hindi: 'Chop the garlic finely'.

Prompts can have **demonstrations** (= examples)

The following are multiple choice questions about high school computer science.

Let $x = 1$. What is $x \ll 3$ in Python 3?

(A) 1 (B) 3 (C) 8 (D) 16

Answer: C

Which is the largest asymptotically?

(A) $O(1)$ (B) $O(n)$ (C) $O(n^2)$ (D) $O(\log(n))$

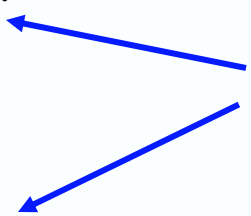
Answer: C

What is the output of the statement `"a" + "ab"` in Python 3?

(A) Error (B) aab (C) ab (D) a ab

Answer:

2 demonstrations



Prompts are a learning signal

This is especially clear with demonstrations

But this is a different kind of learning than pretraining

- Pretraining sets language model weights via gradient descent
- Prompting just changes the context and the activations in the network; no parameters change

We call this **in-context learning**—learning that improves model performance but does not update parameters

LLMs usually have a **system prompt**

<system>You are a helpful and knowledgeable assistant.
Answer concisely and correctly.

This is automatically and silently concatenated to a user prompt

<system> You are a helpful and knowledgeable assistant.
Answer concisely and correctly. **<user>** **What is the capital of France?**

System prompts can be long; 1700 words for Claude Opus4

Some extracts:

Claude should give concise responses to very simple questions, but provide thorough responses to complex and open-ended questions.

Claude is able to explain difficult concepts or ideas clearly. It can also illustrate its explanations with examples, thought experiments, or metaphors.

Claude does not provide information that could be used to make chemical or biological or nuclear weapons.

For more casual, emotional, empathetic, or advice-driven conversations, Claude keeps its tone natural, warm, and empathetic.

Claude cares about people's well-being and avoids encouraging or facilitating self-destructive behavior.

If Claude provides bullet points in its response, it should use markdown, and each bullet point should be at least 1-2 sentences long unless the human requests otherwise.

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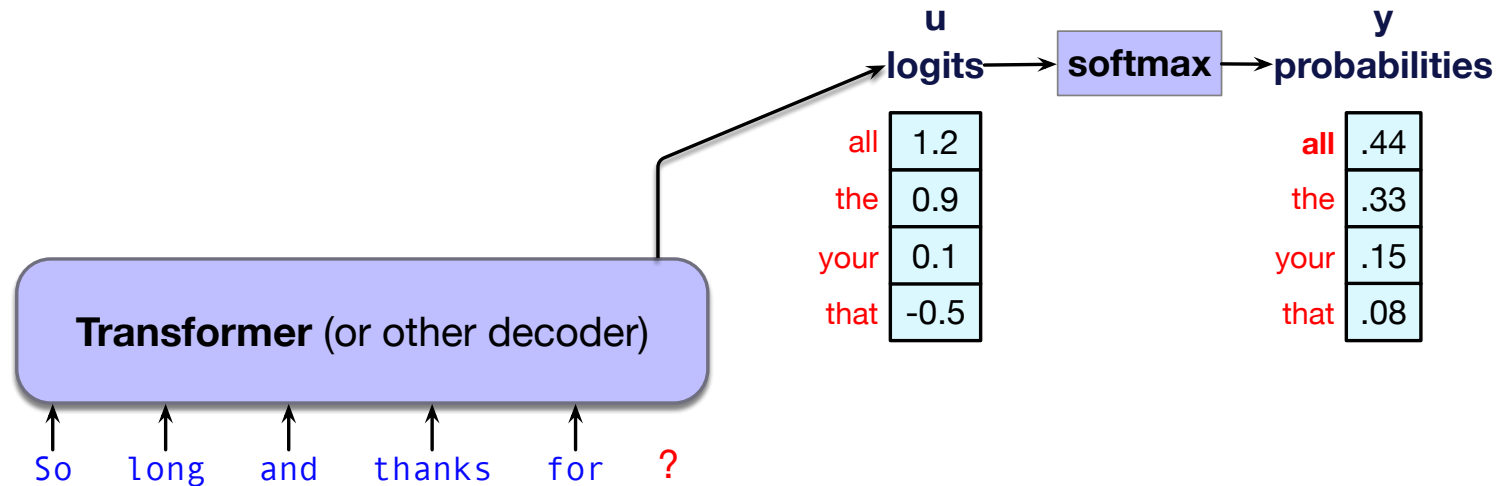
Sampling for LLM Generation

Where does token probability come from?

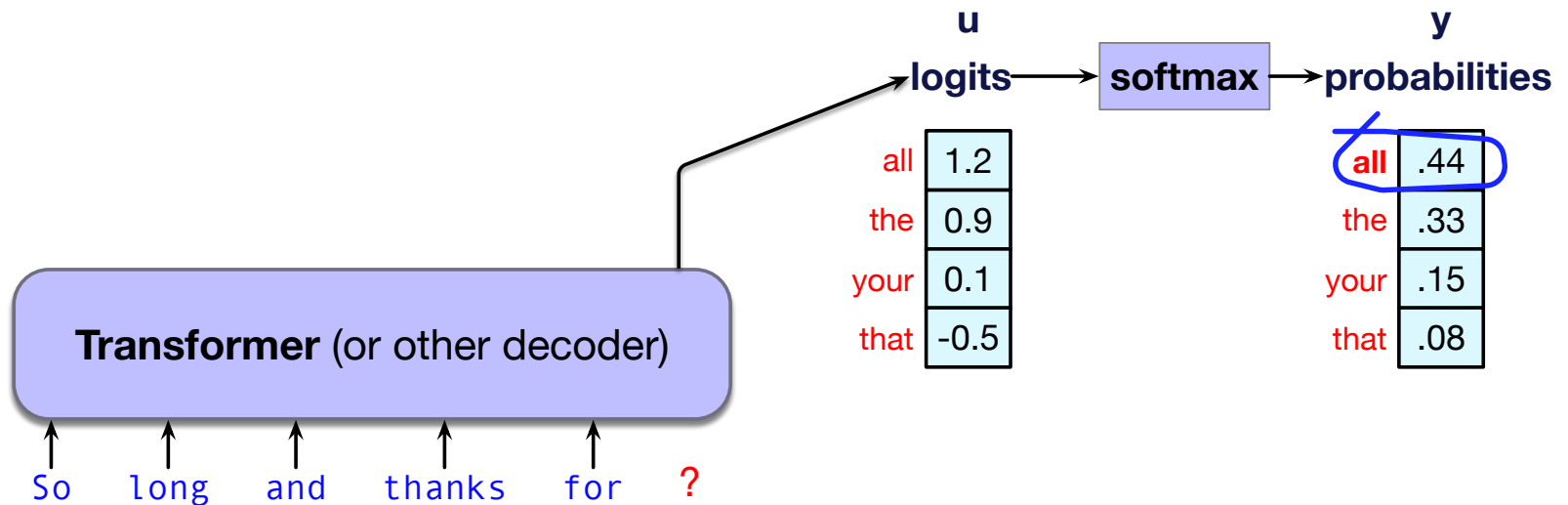
The internal networks for LLMs generate real-valued scores called **logits** for each token in the vocabulary.

Score vector **u** of shape $[1 \times |V|]$ is turned into a probability by softmax

$$\mathbf{y} = \text{softmax}(\mathbf{u})$$



Greedy decoding: choosing "all"





We don't use greedy decoding

Because the tokens it chooses are (by definition) extremely predictable, the resulting text is **generic** and **repetitive**

Greedy decoding is so predictable that it is **deterministic**.

Instead, people prefer text that is more diverse, like that generated by **sampling**

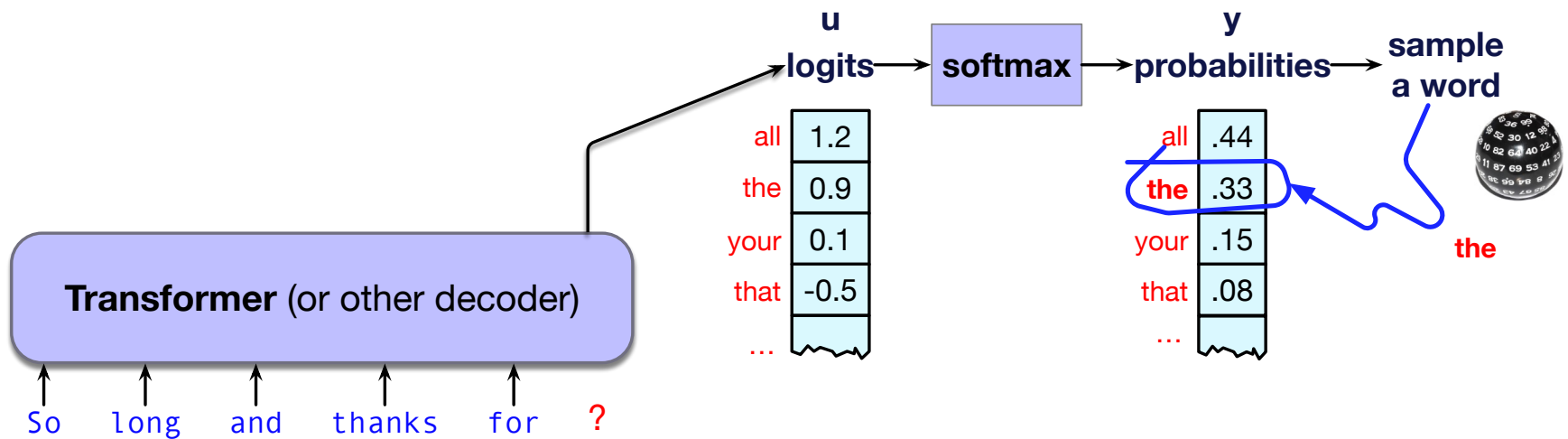
Random sampling

Sampling from a distribution means to choose random points according to their likelihood.

Sampling from an LM means to choose the next token to generate according to its probability.

Random (multinomial) sampling: We randomly select a token to generate according to its probability defined by the LM, conditioned on our previous choices, generate it, and iterate.

Random Sampling





Alas, random sampling doesn't work very well

Even though random sampling mostly generate sensible, high-probable words,

There are many odd, low- probability words in the tail of the distribution

Each one is low- probability but added up they constitute a large portion of the distribution

So they get picked enough to generate weird sentences

Temperature sampling

Divide the logit by a temperature parameter τ before passing it through the softmax.

Instead of

$$\mathbf{y} = \text{softmax}(u)$$

We do

$$\mathbf{y} = \text{softmax}(u/\tau)$$

Temperature sampling

$$0 \leq \tau \leq 1$$

$$\mathbf{y} = \text{softmax}(\mathbf{u}/\tau)$$

Why does this work?

- When τ is close to 1 the distribution doesn't change much.
- The lower τ is, the larger the scores being passed to the softmax
- Softmax pushes high values toward 1 and low values toward 0.
- Large inputs pushes high-probability words higher and low probability word lower, making the distribution more greedy.
- As τ approaches 0, the probability of most likely word approaches 1

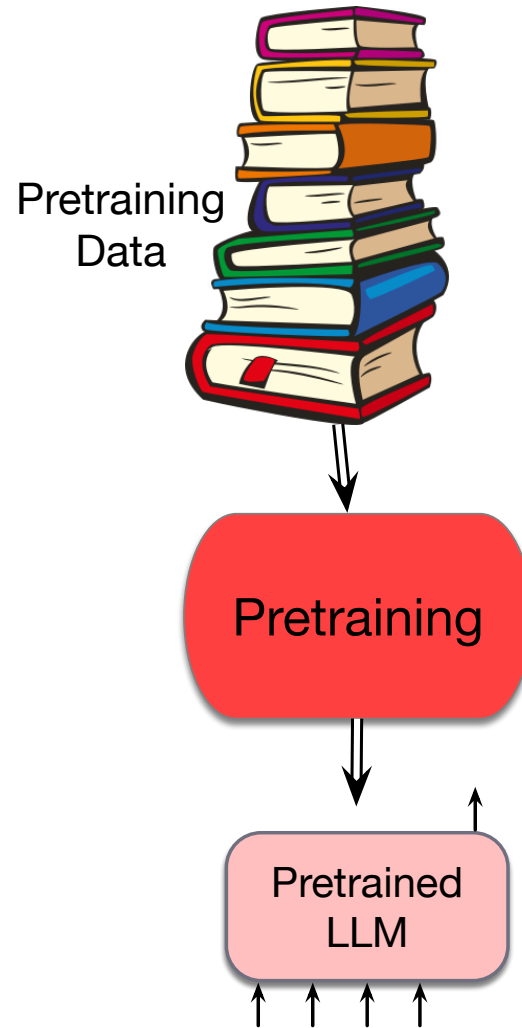
softmax output with temperature τ

logits		close to greedy		normal softmax
		$\tau=0.1$	$\tau=0.5$	$\tau=1$
all	1.2	.95	.59	.44
the	0.9	.05	.32	.33
your	0.1	0	.07	.15
that	-0.5	0	.02	.08

low temperature sampling
(towards greedy)

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Pretraining Large Language Models



Pretraining

The big idea that underlies all the amazing performance of language models

First **pretrain** a transformer model on enormous amounts of text

Then **apply** it to new tasks.

Self-supervised training algorithm

We train them to predict the next word!

1. Take a corpus of text
2. At each time step t
 - i. ask the model to predict the next word
 - ii. train the model using gradient descent to minimize the error in this prediction

"**Self-supervised**" because it just uses the next word as the label!

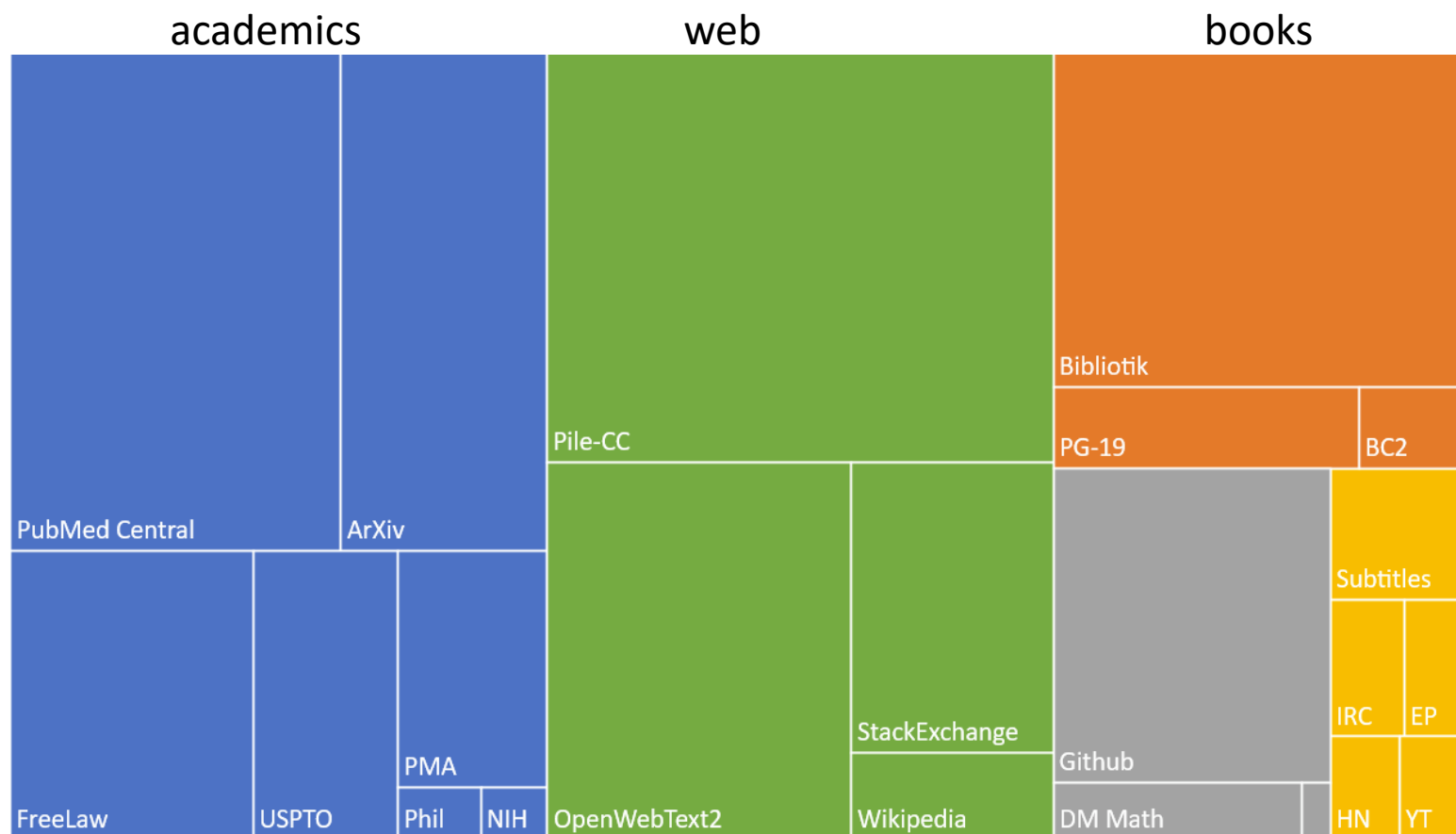
LLMs are mainly trained on the web

Common crawl, snapshots of the entire web produced by the non- profit Common Crawl with billions of pages

Colossal Clean Crawled Corpus (C4; [Raffel et al. 2020](#)), 156 billion tokens of English, filtered

What's in it? Mostly patent text documents, Wikipedia, and news sites

The Pile: a pretraining corpus



dialog

Filtering for quality and safety

Quality is subjective

- Many LLMs attempt to match Wikipedia, books, particular websites
- Need to remove adult content
- Deduplication at many levels (URLs, documents, even lines)

Safety also subjective

- Toxicity detection is important, although that has mixed results
- Can mistakenly flag data written in dialects like African American English

There are problems with scraping from the web



Authors Sue OpenAI Claiming Mass Copyright Infringement of Hundreds of Thousands of Novels

The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work

Millions of articles from The New York Times were used to train chatbots that now compete with it, the lawsuit said.



There are problems with scraping from the web

Copyright: much of the text in these datasets is copyrighted

- Not clear if fair use doctrine in US allows for this use
- This remains an open legal question across the world

Data consent

- Website owners can indicate they don't want their site crawled

Privacy:

- Websites can contain private IP addresses and phone numbers

Skew:

- Training data is disproportionately generated by authors from the US which probably skews resulting topics and opinions

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Finetuning

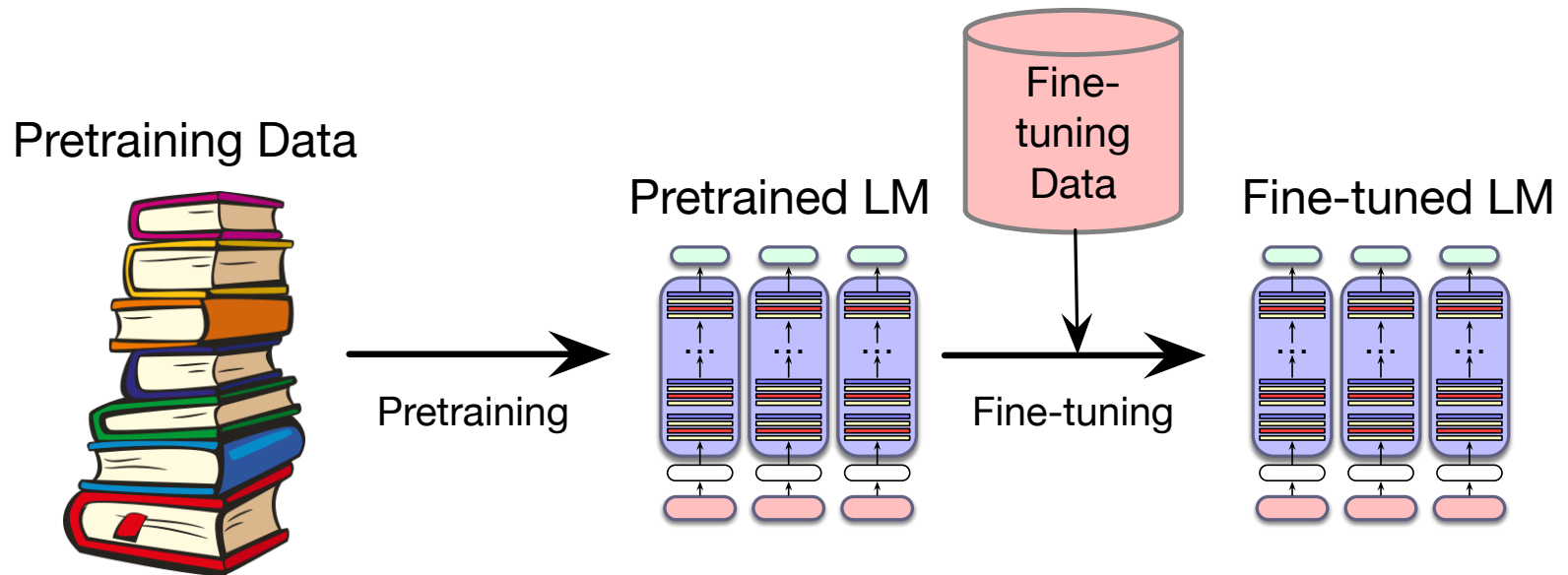
Finetuning for adaptation to new domains

What happens if we need our LLM to work well on a domain it didn't see in pretraining?

Perhaps some specific medical or legal domain?

Or maybe a multilingual LM needs to see more data on some language that was rare in pretraining?

Finetuning



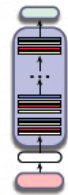
Pretraining

Finetuning

Inference

Finetuning as
Continued
Pretraining

Pretrained LLM



Data from
finetuning
domain

Continue
training all
parameters
on finetuning
domain

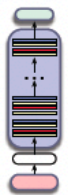
Next word
prediction
objective



On finetuning
domain

Parameter
Efficient
Finetuning
(e.g., LoRA)

Pretrained LLM



Data from
finetuning
domain

Train only new
parameters on
finetuning
domain

Next word
prediction
objective



On finetuning
domain

Task-Specific
Finetuning

Pretrained LLM



Supervised
data from
task

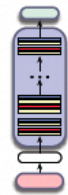
Train only
classification
head on
finetuning
task

Task
specific
loss



On finetuning
task

Instruction
Tuning
(SFT)



Supervised
instructions

Instruction
tuning on
diverse
tasks

Next word
prediction
objective



On unseen
tasks

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Ethical and Safety Issues in Large Language Models

Hallucination

*Chatbots May 'Hallucinate'
More Often Than Many Realize*

What Can You Do When A.I. Lies About You?

People have little protection or recourse when the technology creates and spreads falsehoods about them.

Air Canada loses court case after its chatbot hallucinated fake policies to a customer

The airline argued that the chatbot itself was liable. The court disagreed.

Privacy

How Strangers Got My Email Address From ChatGPT's Model

Abuse and Toxicity

The New AI-Powered Bing Is Threatening Users.

Cleaning Up ChatGPT Takes Heavy Toll on Human Workers

Contractors in Kenya say they were traumatized by effort to screen out descriptions of violence and sexual abuse during run-up to OpenAI's hit chatbot



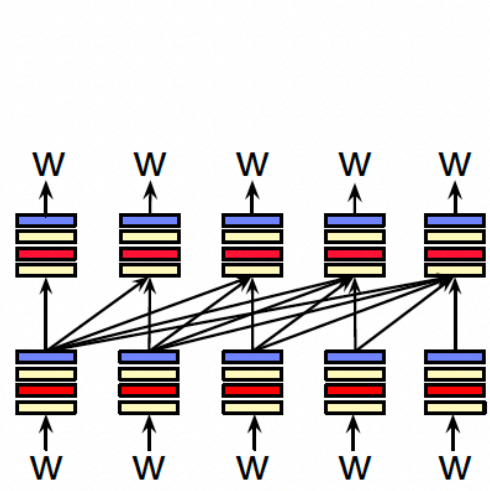
Lots more

Harm (suggesting dangerous actions)

Fraud

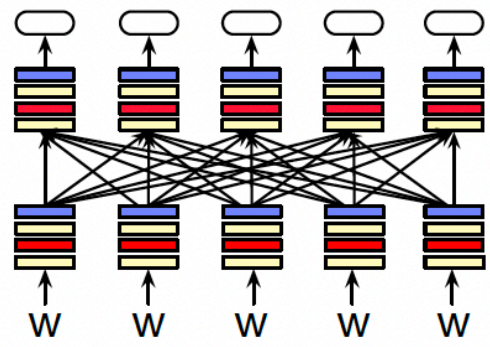
Emotional dependence

Bias



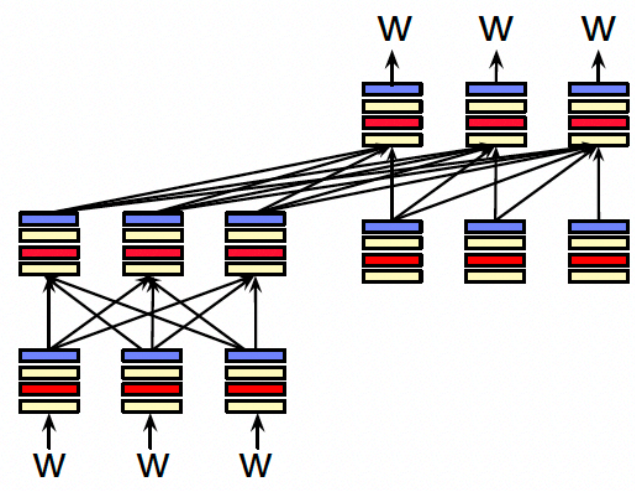
Decoder

Claude
GPT
Mistral
Llama
....



Encoder

BERT
RoBERTa
HuBERT
.....



Encoder-Decoder

BART
T5
MT5
FLAN T5
.....